SELF-DUAL TERNARY CODES OF INDEX 24 IN EVERY LENGTH $12k$

[DRAFT — AUTHORS TO BE ADDED]

Abstract. Gaiotto and Johnson-Freyd attached to every self-dual ternary code C of length n a holomorphic $\mathcal{N} = 1$ superconformal vertex operator algebra of central charge n. Its supersymmetric index is the signed count of full-weight codewords. They conjectured that this index is always divisible by 24, and that for every k some code of length $12k$ has index exactly 24. They found such codes for $k \leq 5$ by computer search, and asked for a systematic construction. We give one. We introduce a block-weighted index $R_\sigma(E)$ attached to a code E and a set σ of disjoint 4-sets of coordinates. We show that Hadamard-gluing tetracodes onto the blocks twice produces a self-dual code of length $n + 12|\sigma|$ whose index is $R_\sigma(E)$, and which again carries a block system with the same value of R. Starting from the extended ternary Golay code, this yields explicit self-dual ternary codes of index 24 in every length $12k$, settling the existence part of the Gaiotto–Johnson-Freyd conjecture.

1. Introduction

A ternary code of length n is a subspace $C \subseteq \mathbb{F}_3^n$; it is *self-dual* if $C = C^\perp$ for the standard form $\langle x, y\rangle = \sum_i x_i y_i$. Self-dual ternary codes exist exactly when $4 \mid n$. A codeword is *maximal* if it has no zero entry, and it is *even* or *odd* according to the parity of the number of entries equal to 1.

Definition 1 ([1, §1])**.** The *index* of a self-dual ternary code C is

$$\mathrm{Index}(C) = \#\{\text{even maximal codewords}\} - \#\{\text{odd maximal codewords}\}.$$

Gaiotto and Johnson-Freyd [1] show that C determines a holomorphic $\mathcal{N} = 1$ SVOA $V_{\Lambda(C)}$ of central charge n, whose Ramond–Ramond partition function (the Witten index) equals $\mathrm{Index}(C)$. The index vanishes unless $12 \mid n$. The Stolz–Teichner programme predicts that an SCFT of central charge $12k$ has index divisible by $24/\gcd(k, 24)$ [3, 1], and suggests that this bound is attained. For codes this leads to:

> **Conjecture** ([1, Conj. 1.4]). The index of every self-dual ternary code is divisible by 24. For every $k \geq 1$ there is a self-dual ternary code of length $12k$ and index 24.

GJF prove the divisibility statement except for lengths $\equiv 12 \pmod{24}$, and exhibit codes of index 24 for $k \leq 5$, found by random search. They ask [1, Question 0.1][1]: *is there a systematic construction of self-dual ternary codes of index* 24 *which would avoid the use of intensive computation?* Our main result answers this question and proves the existence half of the conjecture. The divisibility statement for the remaining lengths is treated separately; here we address only existence.

Theorem 2. *For every $k \geq 1$ there is an explicit self-dual ternary code of length $12k$ and index 24.*

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[1]Numbering of statements in [1] refers to the arXiv version.

Negating one coordinate of a code negates its index, so it suffices to produce index ± 24.

The construction uses only the extended ternary Golay code, tetracodes, and the coordinate-pair map $(x, y) \mapsto (x + y, x - y)$. Its engine is a block-weighted index R_σ (Definition 5). It is preserved by a "partial quadrupling" operation that adds 12 coordinates per block (Lemma 8). Doubling alone multiplies the length by 2 and reaches only $k = 2^j$; the partial version reaches all k.

Conventions. Let $\varphi\colon \mathbb{F}_3 \to \{0, \pm 1\}$ be $\varphi(0) = 0$, $\varphi(1) = -1$, $\varphi(2) = +1$, and $f(x) = \prod_i \varphi(x_i)$ for $x \in \mathbb{F}_3^n$. Then $\mathrm{Index}(C) = \sum_{c \in C} f(c)$. Note $\varphi(-a) = -\varphi(a)$ and $\varphi(a)^2 = 1$ for $a \neq 0$.

2. Two local identities

The *tetracode* $T \subseteq \mathbb{F}_3^4$ is the self-dual $[4, 2, 3]$ code [2], e.g. spanned by $(1, 1, 1, 0)$ and $(0, 1, 2, 1)$. Its nonzero words have weight 3, and each 3-subset of the coordinates is the support of exactly two of them (a word and its negative).

For $a, b \in \mathbb{F}_3$ put $\psi(a, b) = \varphi(a+b)\varphi(a-b)$. Then $\psi(a, b) \neq 0$ iff exactly one of a, b is nonzero, and

$$\psi(a, 0) = 1 \quad (a \neq 0), \qquad \psi(0, b) = -1 \quad (b \neq 0), \tag{1}$$

since $\varphi(a)^2 = 1$ and $\varphi(b)\varphi(-b) = -1$. Similarly let $\kappa(a, b) = \#\{\text{nonzero entries of } (a+b, a-b)\}$. Then $\kappa = 2$ if exactly one of a, b is nonzero, $\kappa = 1$ if both are nonzero (as $b = \pm a$), and $\kappa = 0$ if $a = b = 0$.

Define $\lambda\colon \{0, 1, 2, 3, 4\} \to \mathbb{Z}$ by

$$\lambda(0) = 0, \quad \lambda(1) = -2, \quad \lambda(2) = \lambda(3) = 0, \quad \lambda(4) = 1.$$

Lemma 3 (doubling identity)**.** *For $u \in \mathbb{F}_3^4$,*

$$\sum_{t \in T} \prod_{i=1}^{4} \psi(u_i, t_i) = \lambda(\mathrm{wt}\, u).$$

Proof. A term is nonzero only if $\mathrm{supp}\, t$ is the complement of $\mathrm{supp}\, u$, and then it equals $(-1)^{\mathrm{wt}\, t}$ by (1). Since T has weights 0 and 3 only, with two words on each 3-set, the sum is 1 if $\mathrm{wt}\, u = 4$ ($t = 0$), -2 if $\mathrm{wt}\, u = 1$, and 0 otherwise. □

Lemma 4 (refinement identity)**.** *Split the coordinates of $\mathbb{F}_3^4$ into two pairs $\{p, q\}$ and $\{r, s\}$. For $u \in \mathbb{F}_3^4$,*

$$\sum_{t \in T} \lambda\big(\kappa(u_p, t_p) + \kappa(u_q, t_q)\big)\, \lambda\big(\kappa(u_r, t_r) + \kappa(u_s, t_s)\big) = \lambda(\mathrm{wt}\, u).$$

Proof. Write $L(t)$ for the summand. For $t = 0$ we have $\kappa(u_i, 0) = 2[u_i \neq 0]$, so $L(0) = [\mathrm{wt}\, u = 4]$.

Let $t \neq 0$, so t vanishes at exactly one coordinate z. Let z' be the partner of z in its pair, and $\{x, y\}$ the other pair. On $\{x, y\}$ both $t_x, t_y \neq 0$, so $\kappa(u_x, t_x), \kappa(u_y, t_y) \in \{1, 2\}$. Their sum lies in $\{2, 3, 4\}$, and it is 4 iff $u_x = u_y = 0$; so that factor is $[u_x = u_y = 0]$. On $\{z, z'\}$ the count is $2[u_z \neq 0] + \kappa(u_{z'}, t_{z'})$ with $\kappa(u_{z'}, t_{z'}) \in \{1, 2\}$. It is 4 iff $u_z \neq 0$ and $u_{z'} = 0$, it is 1 iff $u_z = 0$ and $u_{z'} \neq 0$, and otherwise it is 2 or 3. Hence

$$L(t) = [u_x = u_y = 0]\big([u_z \neq 0,\, u_{z'} = 0] - 2\,[u_z = 0,\, u_{z'} \neq 0]\big).$$

This vanishes unless $\mathrm{wt}\, u = 1$. If $\mathrm{supp}\, u = \{w\}$, the two words t vanishing at w contribute $2 \cdot 1$, the two vanishing at the partner of w contribute $2 \cdot (-2)$, and the rest contribute 0. The total is $-2 = \lambda(1)$. If $\mathrm{wt}\, u \neq 1$ only $t = 0$ can contribute, giving $[\mathrm{wt}\, u = 4] = \lambda(\mathrm{wt}\, u)$ (note $\lambda(0) = 0$). □

Both proofs use only the weights of T and the number of words on each 3-set. So Lemmas 3 and 4 hold for every code monomially equivalent to T, in particular for a copy of T placed on any 4-set of coordinates in any order, and Lemma 4 holds for every splitting of the four coordinates into two pairs.

3. Block systems and partial quadrupling

Definition 5. A *block system* σ on a coordinate set X is a set of pairwise disjoint 4-subsets of X (*blocks*). The coordinates of X not in any block are *singletons*. For $x \in \mathbb{F}_3^X$ put

$$w_\sigma(x) = \prod_{i \text{ singleton}} \varphi(x_i) \prod_{\beta\in\sigma} \lambda\big(\mathrm{wt}(x|_\beta)\big), \qquad R_\sigma(C) = \sum_{c\in C} w_\sigma(c).$$

If $\sigma = \emptyset$ then $R_\sigma(C) = \mathrm{Index}(C)$.

Definition 6 (partial quadrupling)**.** Let $E \subseteq \mathbb{F}_3^m$ be self-dual and $\sigma = \{\beta_1, \dots, \beta_b\}$ a block system on $[m]$, with each block split arbitrarily into two pairs. (The splittings enter $Q_\sigma(E)$ but not $R_\sigma(E)$, and by Lemma 4 every choice works below.)

(1) For each coordinate i in a block, add a new coordinate i^*, and place a copy of T on $\beta_j^* = \{i^* : i \in \beta_j\}$ for each j. Apply h_1, the map $(x_i, x_{i^*}) \mapsto (x_i + x_{i^*}, x_i - x_{i^*})$ on every pair $\{i, i^*\}$. For a block $\beta = \{p, q\} \sqcup \{r, s\}$ define the two *step-one blocks* $\{p, p^*, q, q^*\}$ and $\{r, r^*, s, s^*\}$, each again split arbitrarily into two pairs. There are $2b$ of them.

(2) For each coordinate y in a step-one block add a new coordinate y^{**}, and place a copy of T on B^{**} for each step-one block B. Apply h_2, the same map on every pair $\{y, y^{**}\}$.

Set $Q_\sigma(E) = h_2 h_1\big(E \oplus T^b \oplus T^{2b}\big) \subseteq \mathbb{F}_3^{m+12b}$. Singletons of σ are untouched.

Lemma 7. *$Q_\sigma(E)$ is a self-dual code of length $m + 12b$.*

Proof. On one pair, $(x + y)(x' + y') + (x - y)(x' - y') = 2(xx' + yy') = -(xx' + yy')$, so h_1 and h_2 negate the form on the coordinates they pair and fix it elsewhere. Write $v = (e, t_1, t_2)$ with $e \in E$, $t_1 \in T^b$, $t_2 \in T^{2b}$, and split $e = (e_S, e_B)$ into singleton and block parts. Then h_1 negates the form on (e_B, t_1). Next, h_2 negates it on (e_B, t_1, t_2), since those are exactly the coordinates of the step-one blocks and their ** partners. Hence

$$\langle h_2 h_1 v, h_2 h_1 v' \rangle = \langle e, e' \rangle + \langle t_1, t_1' \rangle - \langle t_2, t_2' \rangle = 0$$

for all $v, v' \in E \oplus T^b \oplus T^{2b}$. Since $h_2 h_1$ is bijective, the image has dimension $m/2 + 2b + 4b = (m + 12b)/2$. □

A single step would not suffice: h_1 alone negates the form on the blocks but not on the singletons, and does not preserve self-orthogonality.

Lemma 8 (main lemma)**.** *Let E, σ be as in Definition 6.*

(1) $\mathrm{Index}(Q_\sigma(E)) = R_\sigma(E)$.

(2) *Let σ'' be the block system on the coordinates of $Q_\sigma(E)$ obtained by choosing any subset $\mathcal{B}$ of the step-one blocks, and replacing each $B = \{y_1, y_2\} \sqcup \{y_3, y_4\} \in \mathcal{B}$, split into its two pairs, by the two blocks $\{y_1, y_1^{**}, y_2, y_2^{**}\}$ and $\{y_3, y_3^{**}, y_4, y_4^{**}\}$. All other coordinates are singletons. Then $R_{\sigma''}(Q_\sigma(E)) = R_\sigma(E)$.*

Proof. Part (1) is the case $\mathcal{B} = \emptyset$ of part (2), so we prove (2). Write

$$R_{\sigma''}(Q_\sigma(E)) = \sum_{e,t_1,t_2} w_{\sigma''}\big(h_2 h_1(e, t_1, t_2)\big).$$

Put $u = h_1(e, t_1)$, and sum over $t_2 = (t_B)_B \in T^{2b}$ first. The weight $w_{\sigma''}$ factors over the singletons of σ and over the step-one blocks B together with their ** partners. t_B only enters the factor of B.

Unrefined $B \notin \mathcal{B}$. Its eight coordinates are singletons of σ''. The factor is $\prod_{y\in B} \varphi(u_y + t_{B,y})\varphi(u_y - t_{B,y}) = \prod_{y\in B} \psi(u_y, t_{B,y})$, and summing over $t_B \in T$ gives $\lambda(\operatorname{wt} u|_B)$ by Lemma 3.

Refined $B \in \mathcal{B}$. The factor is

$$\lambda\big(\kappa(u_{y_1}, t_{B,y_1}) + \kappa(u_{y_2}, t_{B,y_2})\big)\, \lambda\big(\kappa(u_{y_3}, t_{B,y_3}) + \kappa(u_{y_4}, t_{B,y_4})\big),$$

and summing over t_B gives $\lambda(\operatorname{wt} u|_B)$ by Lemma 4.

Either way the result is $\sum_{e,t_1} \prod_{i\in S} \varphi(e_i) \prod_B \lambda(\operatorname{wt} u|_B)$, with S the singletons of σ. Now sum over $t_1 = (t_\beta)_\beta \in T^b$. For a block $\beta = \{p, q\} \sqcup \{r, s\}$ the two step-one blocks give

$$\lambda\big(\kappa(e_p, t_{\beta,p}) + \kappa(e_q, t_{\beta,q})\big)\, \lambda\big(\kappa(e_r, t_{\beta,r}) + \kappa(e_s, t_{\beta,s})\big),$$

because $\operatorname{wt} u|_{\{p,p^*,q,q^*\}} = \kappa(e_p, t_{\beta,p}) + \kappa(e_q, t_{\beta,q})$. Lemma 4 turns the sum over t_β into $\lambda(\operatorname{wt} e|_\beta)$. What remains is $\sum_{e\in E} w_\sigma(e) = R_\sigma(E)$. □

Remark 9. If σ has no singletons, Q_σ is two successive Hadamard doublings $E \mapsto H(E \oplus \sigma T)$. Lemma 8 then specialises to the statement that R_σ is preserved under doubling followed by refinement. That gives index-24 codes at lengths $12 \cdot 2^j$ only. The singletons are what make lengths $12k$ with k not a power of 2 reachable.

4. Proof of Theorem 2

Let $G \subseteq \mathbb{F}_3^{12}$ be the extended ternary Golay code [2] with generator matrix

$$\begin{pmatrix}
1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\
0 & 1 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 2 & 2 & 1 \\
0 & 0 & 1 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 2 & 2 \\
0 & 0 & 0 & 1 & 0 & 0 & 2 & 2 & 1 & 0 & 1 & 2 \\
0 & 0 & 0 & 0 & 1 & 0 & 2 & 2 & 2 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 2 & 1 & 2 & 2 & 1 & 0
\end{pmatrix},$$

with coordinates labelled $0, \dots, 11$. $\operatorname{Index}(G) = \pm 24$ (it has 24 maximal codewords, all of the same parity).

Proposition 10. *With $\sigma_1 = \{\{0, 1, 2, 3\}\}$ and $\sigma_2 = \{\{0, 1, 2, 3\}, \{4, 5, 6, 8\}\}$, we have $R_{\sigma_1}(G) = R_{\sigma_2}(G) = 24$.*

Proof. Direct evaluation over the $3^6 = 729$ codewords. More generally, $R_\sigma(G) = \pm 24$ for all 495 single blocks, and for 13,365 of the 17,325 block systems with two blocks; the others give ± 72. □

Lemma 11 (engine)**.** *If E is self-dual of length m and σ is a two-block system with $R_\sigma(E) = \pm 24$, then there is a self-dual E' of length $m + 24$ with $\operatorname{Index}(E') = \pm 24$, and a two-block system σ' on E' with $R_{\sigma'}(E') = \pm 24$.*

Proof. Take $E' = Q_\sigma(E)$, of length $m + 24$ and index $R_\sigma(E)$ by Lemma 8(1). It has four step-one blocks. Refining exactly one of them gives a two-block system σ' with $R_{\sigma'}(E') = R_\sigma(E)$ by Lemma 8(2). □

Proof of Theorem 2. $k = 1$: the Golay code.

Odd $k = 2j + 1 \geq 3$: start from (G, σ_2) and apply Lemma 11 j times. This gives codes of index ± 24 at lengths $12 + 24j$.

Even $k = 2j + 2 \geq 2$: $Q_{\sigma_1}(G)$ has length 24 and index $R_{\sigma_1}(G) = 24$. Refining one of its two step-one blocks gives a two-block system with $R = 24$ (Lemma 8(2)). Applying Lemma 11 j times gives lengths $24 + 24j = 12k$.

Finally, negate one coordinate if necessary to make the index $+24$. □

Remark 12 (Verification)**.** The identities of Lemmas 3, 4 and 8 were also checked by exhaustive enumeration. At length 24 the equality $\mathrm{Index}(Q_\sigma(E)) = R_\sigma(E)$ and invariance under all choices of σ'' hold for 40 random pairs (E, σ) with E of length 12 and $|\sigma| = 1$. The codes produced by the proof at lengths $24, 36, 48, 60, 72, 84$ were constructed explicitly; they are self-dual with $|\operatorname{Index}| = 24$. A maximal codeword is nonzero on an information set, so the $2^{n/2}$ sign patterns on an information set exhaust the candidates ($2^{n/2-1}$ up to $x \mapsto -x$). Enumerating them without sign normalisation gives Index $= +24$ exactly at $n = 24$ and $n = 36$; for $n \leq 84$ the same enumeration, with the sign of one word normalised, gives $|\operatorname{Index}| = 24$. Lengths 72 and 84 are covered neither by the examples of [1] ($k \leq 5$) nor by doubling ($k = 2^j$). Code and generator matrices are available at [repository].

Remark 13. The codes have minimum distance at least 6: by self-duality, a maximal codeword x agrees with a weight-3 codeword w or with $-w$ on $\operatorname{supp} w$, so one of $x \pm w$ is maximal with the opposite sign, and a weight-3 word forces the index to vanish. By construction the codes are built from G and tetracodes only, and a generator matrix is computable in time polynomial in k. This answers [1, Question 0.1]: no search is needed beyond the finite check of Proposition 10 at length 12. Through [1], it yields for every k a holomorphic $\mathcal{N} = 1$ SVOA of central charge $12k$ and index 24. This realises $24\Delta^k$, which is the TMF-predicted minimal value $\frac{24}{\gcd(k,24)}\Delta^k$ exactly when $\gcd(k, 24) = 1$.

References

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